ROSENDO ADRIAN DE LOS RIOS MORENO

Robotics Engineer | Autonomous Systems · Computer Vision · Embedded Systems

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EDUCATION

B.Eng. in Robotics and Digital Systems Engineering — *Jun 2026*

Tecnologico de Monterrey — Monterrey, Mexico

GPA: 91/100   Relevant Coursework: Advanced Embedded Systems, ROS1/ROS2, Computer Vision, Artificial Intelligence, Autonomous Systems

EXPERIENCE

Research Intern — Robotic Teleoperation Systems — *Aug 2025 – Feb 2026*

AIST Tokyo Waterfront, National Institute of Advanced Industrial Science and Technology — Tokyo, Japan

- Reduced obstacle collision risk by implementing **ESDF-based assistive braking** on a **UR5e** arm, achieving <6 cm safety threshold at **40–70 ms braking latency**, by building a real-time **TSDF/ESDF** volumetric reconstruction pipeline in **ROS2** from Intel RealSense D435 depth data.
- Enabled live operator feedback at **15–18 Hz** with **140–210 ms end-to-end latency** by integrating a CPU **TSDF integrator** (Open3D), **DBSCAN** object clustering, and a **ROS-TCP-Connector** Unity bridge into a single teleoperation stack.

INDUSTRY COLLABORATION PROJECTS

Embedded Systems — Autonomous Tractor Navigation — *Aug – Dec 2024*

John Deere · Socio-Formador Project, Tecnologico de Monterrey — Monterrey, Mexico

- Delivered a functional navigation prototype by developing **C/C++ firmware** for **STM32** microcontrollers under **FreeRTOS**, integrating **IMU** sensors via **SPI/I2C** and motor control via **CAN/UART** across a full-semester industry collaboration.

Robotics Engineer — ROS2 Autonomous Mobile Robot — *Feb – Jun 2024*

Manchester Robotics · Socio-Formador Project, Tecnologico de Monterrey — Monterrey, Mexico

- Achieved real-time autonomous navigation by building a full **ROS2** control architecture in **C++/Python** that fused **YOLO**-trained vision with reactive planning, validated through simulation-based unit testing in **Gazebo**.

PROJECTS

Advanced Oil Pan Tracking & Manipulation — *Feb – May 2025*

- Achieved **2.8 cm position** and **1.8° orientation accuracy** at 450 ms latency by building a **ROS2** 3D perception pipeline using **Open3D** and **ICP** on Kinect v2 point clouds, integrated with an **xArm** for 6-DOF manipulation.

3-DOF Teleoperated Robotic Arm — *Feb – May 2025*

- Reduced control latency to **85 ms** end-to-end with <3° angular error by mapping **MediaPipe** full-body pose estimation to servo joint commands published over **ROS2**.

Vision-Based Autonomous Patrol UAV — Tello Robomaster TT — *Feb – Jun 2025*

- Built a fully autonomous patrol drone with real-time person detection by combining **OpenCV** vision, a multi-threaded flight architecture, and a **state machine** controller via **djitellopy** SDK.

SKILLS

Programming	C++, Python, C, MATLAB, C#
Robotics & Sim	ROS1, ROS2, Gazebo, NVIDIA Isaac Sim, FreeRTOS, djitellopy/Robomaster SDK
Vision & Percep.	OpenCV, YOLO, MediaPipe, Open3D, ICP, point cloud processing
Hardware & Proto.	STM32, Raspberry Pi, xArm, Arduino — CAN, SPI, UART, I2C, TCP/IP
DevOps & Tools	Git, Docker, VS Code, Linux
Languages	Spanish (Native) · English B2 (TOEFL IBT Jan 2025)

CERTIFICATIONS

TOEFL IBT — English B2 Certificate — Jan 2025